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Inventor(s)	Yajima; Hisashi et al.

Control method for chuck device

Abstract

A control method of controlling a chuck device in a drive system for driving the chuck device, comprising opening or closing fingers by opening a switching valve at a first degree of valve opening to supply the fluid to a first cylinder chamber of the chuck device and to discharge the fluid from a second cylinder chamber of the chuck device so that a piston moves in one direction, and stopping the fingers in an intermediate position between a fully open position and a fully closed position by opening the switching valve at a second degree of valve opening to stop supplying the fluid to the first cylinder chamber and to stop discharging the fluid from the second cylinder chamber when the axial position of the piston reaches a target position set in advance while the fingers are being opened or closed.

Inventors: Yajima; Hisashi (Tsukuba, JP), Tsuchiya; Gen (Tsukubamirai, JP), Ishikawa; Masayuki (Toride, JP), Kanazawa; Takehiko (Kashiwa, JP)

Applicant: SMC CORPORATION (Tokyo, JP)

Family ID: 73264408

Assignee: SMC CORPORATION (Tokyo, JP)

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Primary Examiner: Singh; Sunil K

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2019-098747 filed on May 27, 2019, and is a Divisional application of U.S. patent application Ser. No. 16/881,055, filed on May 22, 2020, the contents of both of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to control methods for chuck devices having drive systems and capable of gripping workpieces via fingers that can be opened and closed.

Description of the Related Art

(2) Chuck devices gripping and releasing workpieces under the effect of supply of air have been used for, for example, carrying the workpieces. As disclosed in Japanese Patent No. 3932392, the inventors have proposed a chuck device capable of gripping and releasing workpieces by opening and closing gripping members as a piston disposed inside a chuck body is displaced under the effect of supply of compressed air.

(3) In the chuck device, when compressed air is supplied to the inside of the chuck body to move the piston toward an end member, levers engaged with the leading end of the piston rotate, and the gripping members engaged with the leading ends of the respective levers approach each other. As a result, the chuck device enters a gripping state where the chuck device can grip a workpiece. At this moment, the displacement of the piston is restricted by the piston brought into contact with the end member, and the gripping members are kept at a predetermined gripping position.

(4) On the other hand, when compressed air is supplied to a space on a side opposite to the above in the chuck body, the piston moves in a direction away from the end member, and thereby the levers rotate in directions opposite to the above. As a result, the gripping members are separated from each other and enters a release state.

(5) Moreover, in the chuck device, the end member is screwed into the upper end of the chuck body, and the position of the end member along the axial direction can be adjusted. To adjust the gripping position of the gripping members, the axial position of the end member is adjusted by rotating the end member. This limits the amount of stroke of the piston along the axial direction and thereby adjusts the amount of movement of the gripping members in directions along which the gripping members approach each other. In this manner, the gripping position is adjusted from a predetermined position.

SUMMARY OF THE INVENTION

(6) However, in the above-described chuck device, the adjustable amount of movement of the gripping members (the adjustable amount of stroke of the piston) to adjust the gripping position is limited to the movable distance of the end member along the axial direction. In addition, since an operator performs the adjustment by rotating the end member, the adjustment work is complicated, and the positioning with high precision is difficult to achieve. Moreover, there is a demand to make adjustments to the amount of movement of the gripping members and to select the open or closed state of the gripping members using an electrical signal from a higher level controller.

(7) A general object of the present invention is to provide a drive system and a control method for a chuck device capable of stopping fingers serving as gripping members at any position easily and with high precision.

(8) According to an aspect of the present invention, provided is a drive system for driving a chuck device including a piston displaceable in an axial direction under an effect of supply of fluid and a pair of fingers configured to be opened and closed according to the displacement of the piston, the drive system comprising: a controller; a detection sensor configured to detect an axial position of the piston; and a switching valve configured to switch a state of supply of the fluid to the chuck device, wherein the switching valve is a servo valve, of which degree of valve opening is controllable based on a control signal input from the controller, and an amount of displacement of the piston and an amount of opening and closing of the fingers are controlled by changing the degree of valve opening.

(9) According to the present invention, the chuck device includes the piston displaced in the axial direction by the fluid supplied to the chuck device and the pair of fingers opened and closed according to the displacement of the piston. The drive system including the chuck device includes the detection sensor configured to detect the axial position of the piston and the switching valve configured to switch the state of supply of the fluid to the chuck device. The switching valve is a servo valve of which degree of valve opening can be controlled based on the control signal input from the controller, and the amount of displacement of the piston and the amount of opening and closing of the fingers can be controlled by changing the degree of valve opening.

(10) Thus, the degree of valve opening can be controlled by inputting the control signal to the switching valve made of the servo valve, and the direction and amount of displacement of the piston can be controlled by switching the state of supply of the fluid to the chuck device. Consequently, the amount of opening and closing of the pair of fingers can be easily controlled.

(11) As a result, the fingers can be stopped at any position easily and with high precision by freely switching the switching valve using the control signal from the controller compared with the known chuck device that requires the operator to move the end member provided for the chuck body in the axial direction by rotating and thereby to adjust the amount of stroke of the piston so that the gripping members are stopped at a freely selected position.

(12) The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic circuit diagram illustrating a drive system for a chuck device according to an embodiment of the present invention;

(2) FIG. 2 is an overall cross-sectional view of the chuck device constituting the drive system in FIG. 1;

(3) FIG. 3 is a schematic circuit diagram of the drive system in FIG. 1 when a switching valve is switched from a first switch position to a second switch position to change a gripping portion to a release state;

(4) FIG. 4 is a schematic circuit diagram of the drive system in FIG. 1 when the switching valve is switched to a third switch position to change the gripping portion to a gripping state;

(5) FIG. 5 is a diagram of characteristic curves illustrating the relationships between a valve command signal to the switching valve and the supply rate, the discharge rate, and the valve opening in the drive system in FIG. 1;

(6) FIG. 6 is a time chart illustrating the relationships between time and a command signal to a controller, the valve command signal from the controller to the switching valve, and a piston position signal in the drive system in FIG. 1;

- (7) FIG. 7 is a schematic circuit diagram of a drive system obtained by providing a pair of pressure sensors capable of detecting the pressures in a head-side cylinder chamber and a rod-side cylinder chamber for the drive system in FIG. 1;
- (8) FIG. 8 is a schematic circuit diagram of a drive system obtained by providing a differential pressure sensor capable of detecting the difference between the pressure in the head-side cylinder chamber and the pressure in the rod-side cylinder chamber for the drive system in FIG. 1; and
- (9) FIG. 9 is a schematic circuit diagram illustrating a drive system for the chuck device according to a modification, the drive system including a pair of switching valves.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (10) As illustrated in FIG. 1, a drive system **10** for driving a chuck device **12** according to an embodiment of the present invention is applied to the chuck device **12** opening and closing under the effect of supply of compressed air and includes a switching valve **14** switching a state of supply of compressed air (fluid) to the chuck device **12** and a controller **16** outputting a valve command signal (control signal) to the switching valve **14** based on an input command signal. Moreover, the drive system **10** selectively connects first and second ports **66** and **70** (described below) to first and second body ports **36** and **38** (described below), respectively, of the chuck device **12** through the switching action of the switching valve **14**. In addition, the controller **16** stores a control map for driving the chuck device **12** in advance.
- (11) First, the configuration of the chuck device **12** will be described with reference to FIGS. 1 and 2.
- (12) The chuck device **12** includes a tubular chuck body **18**, a piston **20** disposed inside the chuck body **18** to be freely displaced, a gripping portion **30** including a pair of first and second fingers **26** and **28** that open and close via levers **24a** and **24b**, respectively, engaged with a rod part **22** of the piston **20**, and a detection sensor **32** disposed on a side part of the chuck body **18** to detect the position of the piston **20**. The chuck device **12** described above is an air chuck opening and closing under the effect of supply of compressed air.
- (13) The chuck body **18** has, for example, a substantially rectangular cross-section. The chuck body **18** has a body hole **34** passing through the chuck body **18** in the axial direction (directions of arrows A and B) and the first and second body ports **36** and **38** opened in a side surface and communicating with the body hole **34** to supply and discharge pressurized fluid.
- (14) The body hole **34** is opened in one end part of the chuck body **18** and closed by a head cover **40** fitted in the body hole **34**. The head cover **40** is held by a stop ring **42** while being engaged with a stepped part of the body hole **34**, and the movement of the head cover **40** toward the open one end part (in the direction of the arrow A) is restricted by the stop ring **42**.
- (15) The body hole **34** is partitioned into a head-side cylinder chamber (cylinder chamber) **44** (described below) formed between the piston **20** and the head cover **40** and a rod-side cylinder chamber (cylinder chamber) **46** defined between the piston **20** and another end part of the chuck body **18**.
- (16) The first and second body ports **36** and **38** are separate from each other in the axial direction of the chuck body **18** (in the directions of the arrows A and B) by a predetermined distance. The first body port **36** communicates with the head-side cylinder chamber **44**, and the second body port **38** communicates with the rod-side cylinder chamber **46**. The first and second body ports **36** and **38** are connected to the first and second ports **66** and **70** (described below) of the switching valve **14** via first and second tubes **68** and **72**, respectively, so that compressed air is supplied and discharged through the switching action of the switching valve **14**.
- (17) On the other hand, a plate-like base body **48** is joined to the other end part of the chuck body **18** to be substantially orthogonal to the axis of the chuck body **18** and thus closes the other end part. The base body **48** has a pair of insertion holes **49a** and **49b** in which the levers **24a** and **24b** are partially fitted.
- (18) Moreover, a rail groove **50** is formed in the base body **48** so that the pair of first and second

fingers **26** and **28** constituting the gripping portion **30** are guided to be freely displaced along the rail groove **50**. The rail groove **50** is formed in a side surface of the base body **48** on a side opposite the side on which the chuck body **18** lies, and linearly extends in a longitudinal direction (in a direction of an arrow C in FIG. 2) substantially orthogonal to the axis of the chuck body **18**.

(19) The piston **20** can be displaced in the axial direction (in the directions of the arrows A and B) such that, for example, the outer circumferential surface can slide along the inner circumferential surface of the body hole **34**. A piston packing **52** attached to the outer circumferential surface is in contact with the inner circumferential surface, and a magnet **54** is disposed adjacent to the piston packing **52**.

(20) The detection sensor **32** attached to the side surface of the chuck body **18** detects the magnetism of the magnet **54** and outputs the results in the form of an electrical signal to the controller **16**. The axial position and the displacement speed of the piston **20** along the axial direction (along the directions of the arrows A and B) are calculated from the output.

(21) Moreover, one end face of the piston **20** faces the head cover **40** inside the body hole **34**, and another end face is provided with the rod part **22** having a reduced diameter and extending toward the base body **48** (in the direction of the arrow B). A damper **55** composed of an elastic material is attached to a border region between the rod part **22** and the other end face. The rod part **22** is fitted in a rod hole **56** having a smaller diameter than the body hole **34**, and the leading end is provided with a link pin **58** orthogonal to the axial direction of the rod part **22**. The pair of levers **24a** and **24b** are rotatably engaged with the leading end of the rod part **22** via the link pin **58**.

(22) The levers **24a** and **24b** having a substantially L-shaped cross-section are paired with each other and disposed to be symmetric with respect to the axis of the piston **20**. The levers **24a** and **24b** are rotatably supported by support shafts **60a** and **60b** provided for the chuck body **18** and respectively fitted in bent, substantially middle parts of the levers **24a** and **24b**.

(23) Moreover, first end parts of the levers **24a** and **24b** respectively engaged with the link pin **58** via semicircular notches **25a** and **25b**, and second end parts **27a** and **27b** having a deformed cross-section and expanding into a spherical shape are respectively fitted and held in engagement holes **61a** and **61b** in the first and second fingers **26** and **28** constituting the gripping portion **30**.

(24) That is, the first end parts of the pair of levers **24a** and **24b** move in the axial direction under the effect of displacement of the piston **20** (the rod part **22**) along the axial direction (along the directions of the arrows A and B) via the link pin **58**, and as a result of this, the second end parts **27a** and **27b** respectively rotate on the support shafts **60a** and **60b** to approach or be separated from each other.

(25) The gripping portion **30** includes the pair of first and second fingers (fingers) **26** and **28** capable of being displaced along the rail groove **50** of the base body **48**. The first and second fingers **26** and **28** respectively include block-like body parts **62a** and **62b** guided along the rail groove **50** and claw parts **64a** and **64b** respectively protruding from the body parts **62a** and **62b** to be substantially orthogonal to the body parts to grip a workpiece.

(26) The first and second fingers **26** and **28** are disposed to be movable to approach and be separated from each other along the rail groove **50** of the base body **48**. In this case, the first finger **26** and the second finger **28** move symmetrically with respect to the axis of the piston **20**.

(27) Next, the configuration of the switching valve **14** will be described with reference to FIGS. 1, 3, and 4.

(28) The switching valve **14** is, for example, a 5-port servo valve including a valve element (not illustrated) that opens and closes in response to the valve command signal from the controller **16**. The first port **66** is connected to the first body port **36** of the chuck device **12** via the first tube **68**, and the second port **70** is connected to the second body port **38** via the second tube **72**.

(29) Moreover, in the switching valve **14**, a supply port **74** is connected to an air supply source (supply source) **78** supplying compressed air via a third tube **76**, and first and second exhaust ports **80** and **82** communicate with the outside.

(30) When the switching valve **14** is in a first switch position **P1** as illustrated in FIG. **1**, the first and second ports **66** and **70** are not connected to any of the supply port **74** and the first and second exhaust ports **80** and **82**. Thus, the switching valve **14** blocks supply of compressed air from the air supply source **78** to the chuck device **12** and discharge of compressed air from the chuck device **12**, bringing the chuck device **12** to a standstill.

(31) When the switching valve **14** is switched to a second switch position **P2** as illustrated in FIG. **3** in response to the valve command signal, the supply port **74** communicates with the first port **66**, and thus the air supply source **78** connected to the supply port **74** communicates with the first body port **36** of the chuck device **12**. As a result, compressed air is supplied to the head-side cylinder chamber **44**. At the same time, the second port **70** communicates with the second exhaust port **82**, and thus the second body port **38** of the chuck device **12** communicates with the outside.

(32) When the switching valve **14** is switched to a third switch position **P3** as illustrated in FIG. **4** in response to the valve command signal, the first port **66** communicates with the first exhaust port **80**, and thus the first body port **36** of the chuck device **12** communicates with the outside. At the same time, the supply port **74** communicates with the second port **70**. As a result, the air supply source **78** communicates with the second body port **38** of the chuck device **12**, and compressed air is supplied to the rod-side cylinder chamber **46**.

(33) That is, the switching valve **14** described above can be freely and sequentially switched between the first switch position **P1**, the second switch position **P2**, and the third switch position **P3** in response to the valve command signal input from the controller **16** based on, for example, a command voltage serving as the command signal. Specifically, the switching valve **14** is switched to the second switch position **P2**, the first switch position **P1**, and the third switch position **P3** in this order as the value of the valve command signal gradually increases as illustrated in FIG. **5**. The valve command signal is a signal for inputting, for example, timings of gripping or releasing and the amount of opening (amount of movement) of the first and second fingers **26** and **28** to the controller **16**.

(34) The drive system **10** for the chuck device **12** according to the embodiment of the present invention is basically configured as above. Next, the operations and operational effects thereof will be described. In the description below, a state where the switching valve **14** is in the first switch position **P1** as illustrated in FIG. **1** is defined as an initial state.

(35) To change the gripping portion **30** of the chuck device **12** from the initial state to a release state (open), the command signal indicating to change the gripping portion **30** to the open state is input to the controller **16**, and the valve command signal is input from the controller **16** to the switching valve **14** to move the valve element (not illustrated). This causes the switching valve **14** to be switched to the second switch position **P2** in which the supply port **74** is connected to the first port **66** while the second port **70** is connected to the second exhaust port **82** as illustrated in FIG. **3**.

(36) As a result, compressed air from the air supply source **78** is supplied to the first body port **36** of the chuck device **12** through the first tube **68** after flowing through the third tube **76**, and the supply port **74** and the first port **66** of the switching valve **14**.

(37) Then, the piston **20** of the chuck device **12** is pushed by the compressed air supplied from the first body port **36** to the head-side cylinder chamber **44** and moves toward the other end part (in the direction of the arrow **B**) along the chuck body **18**. As the piston **20** moves, the compressed air in the rod-side cylinder chamber **46** is discharged from the second body port **38** through the second tube **72** and from the second port **70** of the switching valve **14** to the outside through the second exhaust port **82**.

(38) The movement of the piston **20** toward the other end part (in the direction of the arrow **B**) causes the pair of levers **24a** and **24b** engaged with the rod part **22** to rotate respectively on the support shafts **60a** and **60b** such that the second end parts **27a** and **27b** are separated from each other. As a result, the first and second fingers **26** and **28** move outward along the rail groove **50** to be separated from each other, and enter the release state.

(39) Moreover, the magnetism of the magnet **54** attached to the piston **20** is detected by the detection sensor **32** disposed on the chuck body **18** and is output to the controller **16** as a position signal. This enables the axial position of the piston **20** along the axial direction to be determined. Thus, based on the position of the piston **20**, it is determined that the first and second fingers **26** and **28** are in an open position (release state; open) where the first and second fingers **26** and **28** are separated from each other.

(40) In this case, the piston **20** is displaced until coming into contact with an end part of the body hole **34**, and then stops. The amount of outward movement (the amount of opening) of the first and second fingers **26** and **28** corresponds to the axial position of the piston **20**.

(41) Next, how the gripping portion **30** including the first and second fingers **26** and **28** in the above-described release state is switched to a gripping state where the gripping portion **30** grips a workpiece or the like will be described.

(42) First, the command signal indicating to change the gripping portion **30** to a closed state is input to the controller **16**, and the valve command signal is input from the controller **16** to the switching valve **14** to move the valve element (not illustrated). This causes the switching valve **14** to be switched to the third switch position P3 in which the supply port **74** is connected to the second port **70** while the first port **66** is connected to the first exhaust port **80** as illustrated in FIG. 4.

(43) As a result, supply of compressed air from the air supply source **78** to the first body port **36** stops while the compressed air is supplied to the second body port **38** of the chuck device **12** through the second tube **72** after flowing through the supply port **74** and the second port **70** of the switching valve **14**.

(44) Then, the piston **20** of the chuck device **12** is pushed by the compressed air supplied from the second body port **38** to the rod-side cylinder chamber **46** and moves toward the one end part (in the direction of the arrow A) along the chuck body **18**. As the piston **20** moves, the compressed air in the head-side cylinder chamber **44** is discharged from the first body port **36** through the first tube **68** and from the first port **66** of the switching valve **14** to the outside through the first exhaust port **80**.

(45) The movement of the piston **20** toward the one end part (in the direction of the arrow A) causes the pair of levers **24a** and **24b** to rotate respectively on the support shafts **60a** and **60b** such that the second end parts **27a** and **27b** approach each other. As a result, the first and second fingers **26** and **28** move inward along the rail groove **50** to approach each other. Thus, the gripping portion **30** enters the gripping state where the gripping portion **30** can grip a workpiece (not illustrated) disposed between the claw part **64a** of the first finger **26** and the claw part **64b** of the second finger **28**.

(46) Moreover, the magnetism of the magnet **54** attached to the piston **20** is detected by the detection sensor **32** and is output to the controller **16** as the position signal. As a result of this, the axial position of the piston **20** is determined, and at the same time, it is determined that the first and second fingers **26** and **28** are in a closed position (gripping state; closed) where the first and second fingers **26** and **28** are close to each other.

(47) In this case, the piston **20** moves toward the head cover **40** until the first and second fingers **26** and **28** grip the workpiece or until the first and second fingers **26** and **28** come into contact with each other.

(48) Next, how the gripping portion **30** including the first and second fingers **26** and **28** is stopped at a freely selected intermediate position between a release position and a gripping position will be described with reference to FIGS. 5 and 6. In the description below, the first and second fingers **26** and **28** are moved from the closed position (gripping state) and stopped in the intermediate position.

(49) FIG. 5 described above is a diagram of characteristic curves illustrating the relationships between the valve command signal from the controller **16** to the switching valve **14** and the state of

supply and discharge of compressed air at the first and second ports **66** and **70**. In FIG. 5, a characteristic curve **F1** drawn with a solid line illustrates the state of supply and discharge of compressed air to and from the first port **66** (the first body port **36**), and a characteristic curve **F2** drawn with a broken line illustrates the state of supply and discharge of the compressed air to and from the second port **70** (the second body port **38**). That is, the state of supply and discharge of compressed air to and from the first port **66** and the flow rate (opening) change according to the value of the valve command signal. The same applies to the second port **70**.

(50) First, the command signal indicating to change the gripping portion **30** to the closed state is input to the controller **16**, and the valve command signal indicating to open the switching valve **14** at a first opening V_a is input to the switching valve **14** based on the command signal. This causes the switching valve **14** to be switched to the third switch position **P3** and thus causes the first and second fingers **26** and **28** to be placed in the closed position. The axial position of the piston **20** is detected by the detection sensor **32** and is input to the controller **16** as the position signal. As a result, it is determined that the first and second fingers **26** and **28** are in the closed position.

(51) As can be understood from FIG. 5, in the third switch position **P3**, the opening of the first port **66** and the opening of the second port **70** at a predetermined value of the valve command signal are identical to each other, and the flow rate of compressed air discharged from the chuck device **12** through the first port **66** (the characteristic curve **F1**) and the flow rate of compressed air supplied to the chuck device **12** through the second port **70** (the characteristic curve **F2**) are substantially identical to each other.

(52) Next, the command signal indicating to change the gripping portion **30** from the closed state to the release state is input to the controller **16**, and the valve command signal based on the command signal is input from the controller **16** to the switching valve **14**. This causes the switching valve **14** to be switched to the second switch position **P2** (see FIG. 3) in which the switching valve **14** is opened at a second opening (first degree of valve opening) V_b . As a result, as illustrated in FIG. 3, compressed air from the air supply source **78** is supplied to the first body port **36** through the supply port **74** and the first port **66** while the compressed air inside the rod-side cylinder chamber **46** is discharged from the second port **70** through the second exhaust port **82**.

(53) Also in the second switch position **P2**, the opening of the first port **66** and the opening of the second port **70** at a predetermined value of the valve command signal are identical to each other, and the flow rate of compressed air supplied to the chuck device **12** through the first port **66** (the characteristic curve **F1**) and the flow rate of compressed air discharged from the chuck device **12** through the second port **70** (the characteristic curve **F2**) are substantially identical to each other.

(54) This causes the piston **20** of the chuck device **12** to start moving in a direction away from the head cover **40** (in the direction of the arrow **B**), and as a result, causes the first and second fingers **26** and **28** to start moving along the rail groove **50** to be separated from each other.

(55) The change in the axial position of the piston **20** is detected by the detection sensor **32**. When the detection sensor **32** detects that the piston **20** reaches a switch point **E**, which is a predetermined distance **L** short of a target position **D** of the piston **20** corresponding to the intermediate position of the first and second fingers **26** and **28**, the valve command signal indicating to open the switching valve **14** at a third opening (third degree of valve opening) V_c is output from the controller **16** to the switching valve **14**.

(56) The third opening V_c is a degree of opening at which the switching valve **14** is switched from the second switch position **P2** to the third switch position **P3**, and compressed air is supplied from the second body port **38** to the rod-side cylinder chamber **46** while the compressed air in the head-side cylinder chamber **44** is discharged through the first body port **36**. As illustrated in FIG. 6, the third opening V_c is set closer to the first opening V_a than a degree of opening at which both the rod side and the head side are blocked.

(57) After the switching valve **14** is opened at the third opening V_c for a predetermined period t , the switching valve **14** is switched to be opened at a fourth opening (second degree of valve

opening) Vd in response to the updated valve command signal from the controller **16**. This causes the switching valve **14** to be switched to the first switch position P1 (see FIG. **1**) in which the supply and discharge of compressed air to and from the first and second body ports **36** and **38** are stopped.

(58) Moreover, the axial position of the piston **20** is input to the controller **16** as the position signal, whereby it is determined that the first and second fingers **26** and **28** are open and at a standstill in the freely selected intermediate position.

(59) That is, the switching valve **14** is switched to stop supplying compressed air to the first body port **36** and to supply compressed air to the second body port **38** for the predetermined period t before the first and second fingers **26** and **28** reach the predetermined intermediate position (the target position D of the piston **20**). As a result, the displacement speed of the piston **20** can be favorably reduced, and the piston **20** can be stopped at the predetermined axial position (the target position D) with high precision. Thus, the first and second fingers **26** and **28** can be stopped in the desired intermediate position with high precision.

(60) Moreover, the displacement speed of the first and second fingers **26** and **28** is set by the first opening Va and the second opening Vb of the switching valve **14**, and the precision in stopping the first and second fingers **26** and **28** in the intermediate position is determined by the third opening Vc, the predetermined distance L to the target position D of the piston **20** along the axial direction, and the predetermined period t in which the switching valve **14** is opened at the third opening Vc.

(61) The optimum values of the first opening Va, the second opening Vb, the third opening Vc, the predetermined distance L to the target position D of the piston **20**, and the predetermined period t vary depending on various conditions of the chuck device **12** such as the inner diameter of the body hole **34** in the chuck body **18** and the amount of stroke of the piston **20** along the axial direction, and are stored in the control map of the controller **16** in advance according to the specifications and the driving conditions of the chuck device **12**.

(62) Moreover, the position of the piston **20** detected by the detection sensor **32** is output to the controller **16**, and the valve command signal output to the switching valve **14** is changed based on the axial position and the displacement speed of the piston **20** calculated by the controller **16** to change the valve opening of the switching valve **14** as appropriate. As a result, opening and closing of the first and second fingers **26** and **28** can be controlled more precisely.

(63) Furthermore, as in a drive system **90** illustrated in FIG. **7**, the drive system may be provided with pressure sensors **92a** and **92b** respectively capable of detecting the pressures inside the head-side cylinder chamber **44** (the first body port **36**) and the rod-side cylinder chamber **46** (the second body port **38**) in the chuck device **12**. At least one of the pressures in the head-side cylinder chamber **44** and the rod-side cylinder chamber **46** respectively detected by the pressure sensors **92a** and **92b** may be output to the controller **16**, and the valve command signal based on the pressures may be output to the switching valve **14** to set the valve opening and the switching timing of the switching valve **14**.

(64) Yet moreover, as in a drive system **100** illustrated in FIG. **8**, the drive system may be provided with a differential pressure sensor **102** capable of detecting the difference between the pressure in the head-side cylinder chamber **44** (the first body port **36**) and the pressure in the rod-side cylinder chamber **46** (the second body port **38**). The pressure difference detected by the differential pressure sensor **102** may be output to the controller **16**, and the valve command signal based on the pressure difference may be output to the switching valve **14** to set the valve opening and the switching timing of the switching valve **14**.

(65) In this manner, the drive system **90** provided with the pressure sensors **92a** and **92b** and the drive system **100** provided with the differential pressure sensor **102** can precisely control the opening and closing of the first and second fingers **26** and **28** and, at the same time, can favorably adjust the gripping force of the first and second fingers **26** and **28** during gripping of the workpiece based on the pressures detected by the pressure sensors **92a** and **92b** and the pressure difference

detected by the differential pressure sensor **102**, respectively.

(66) Moreover, in a case where the volumes of the head-side cylinder chamber **44** and the rod-side cylinder chamber **46** of the chuck device **12** are relatively small, supply and discharge of compressed air need to be switched quickly in a short time. Thus, for example, a 5-port air servo valve, which directly drives a spool (valve element) in the axial direction using a linear motor, is the most suitable for the switching valve **14**.

(67) On the other hand, the switching valve **14** switching the state of supply of compressed air to the chuck device **12** is not limited to the 5-port servo valve as in the above-described drive systems **10**, **90**, and **100**. For example, as in a drive system **110** illustrated in FIG. **9**, the drive system may be provided with a pair of switching valves **112a** and **112b**, which are 3-port servo valves, and an output port **114a** of the switching valve **112a** may be connected to the first body port **36** of the chuck device **12** while an output port **114b** of the switching valve **112b** may be connected to the second body port **38** of the chuck device **12**.

(68) To change the chuck device **12** to the open state in the drive system **110**, compressed air is supplied from the switching valve **112a** to the first body port **36** while compressed air is discharged from the chuck device **12** to the outside through the switching valve **112b**. This causes the first and second fingers **26** and **28** to enter the release state (open) under the effect of displacement of the piston **20**.

(69) On the other hand, to change the chuck device **12** to the closed state, compressed air is supplied from the switching valve **112b** to the second body port **38** while compressed air is discharged from the chuck device **12** to the outside through the switching valve **112a**. This causes the first and second fingers **26** and **28** to enter the gripping state (closed) under the effect of displacement of the piston **20**.

(70) Moreover, the first and second fingers **26** and **28** can also be stopped at a freely selected intermediate position by switching the state of supply of compressed air to the chuck device **12** as appropriate using the switching valves **112a** and **112b**.

(71) The drive systems **10**, **90**, **100**, and **110** and the control methods for the chuck device **12** according to the present invention are not limited in particular to the embodiments described above, and may have various configurations without departing from the scope of the present invention as a matter of course.

Claims

1. A control method of controlling a chuck device in a drive system for driving the chuck device, the chuck device including a piston displaceable in an axial direction under an effect of supply of fluid and a pair of fingers configured to be opened and closed according to the displacement of the piston, the drive system comprising a controller, a detection sensor configured to detect an axial position of the piston, and a switching valve configured to switch a state of supply of the fluid to the chuck device, wherein the switching valve is a servo valve whose degree of valve opening is controllable based on a control signal input from the controller, and an amount of displacement of the piston and an amount of opening and closing of the fingers are controlled by changing the degree of valve opening, the control method comprising the steps of: opening or closing the fingers by opening the switching valve at a first degree of valve opening to supply the fluid to a first cylinder chamber of the chuck device and to discharge the fluid from a second cylinder chamber of the chuck device so that the piston moves in one axial direction; reducing, while the fingers are being opened or closed, a displacement speed of the piston moving in the one axial direction by switching the valve opening to a second degree of valve opening to supply the fluid to the second cylinder chamber and to discharge the fluid from the first cylinder chamber when the detection sensor detects that the piston reaches a switching point, which is ahead of a predetermined target position of the piston in said one axial direction by a predetermined distance, such that the piston

reaches the switching point before reaching the predetermined target position when moving in said one axial direction; and stopping the fingers at an intermediate position located between a fully open position and a fully closed position by opening the switching valve at a third degree of valve opening to stop supplying the fluid to the second cylinder chamber and to stop discharging the fluid from the first cylinder chamber when the piston reaches the predetermined target position.

2. The control method according to claim 1, wherein the switching valve is opened at the second degree of valve opening for a predetermined period and then switched to be opened at the third degree of valve opening.
